Sun 2021 HCC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6 patients (P04, P05, P09, P10, P16, P18) intermix on one connected component; a cycling-enriched lobe (~32%) does not separate, is not gastrulation-high, and leaves the Gast↔Crest gradient intact when removed → full manifold, no scoping.

Gate 2 — module geography → GO (non-branching)

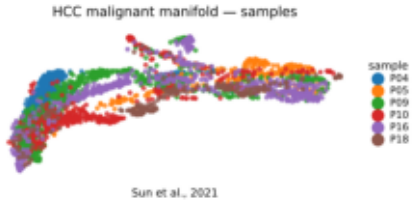
Principal curve fold-free (2 tips, 0 forks); shared developmental gradient — Gast declines (ρ=−0.79) while Crest rises to the terminus (ρ=+0.60); smoothed Gast↔Crest r=−0.71 (inverse terminal gradient).

Gate 3 — hub mutual exclusivity → NA

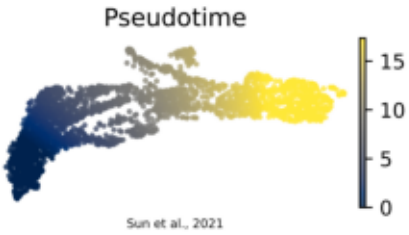
Non-branching manifold — no fork, so the Gast↔pSom cross-repression test does not apply (branching-only gate; §2).

STEP FIGURES

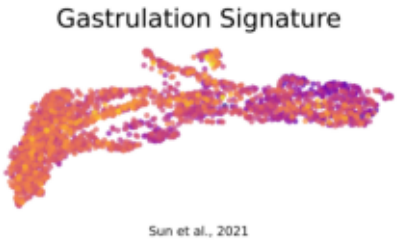
Manifold — samples



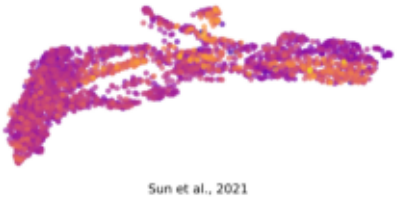
Pseudotime



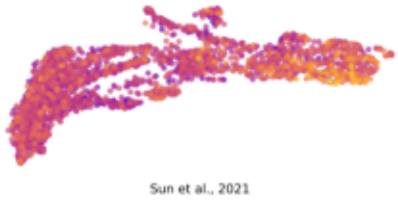
Gastrulation



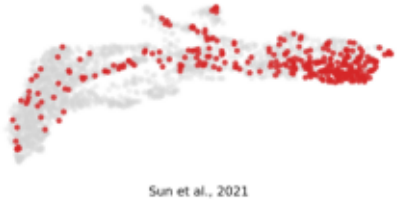
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Hepatocellular carcinoma (Sun et al., 2021), 10x — 16,498 cells / 18 samples (2,737 malignant). Malignant only: 2,338 pass the thin-sample threshold (cut 125, keeps 6 samples); 2,337 enter the trajectory after 1-cell island removal.

Verdict.
Malignant cells recapitulate an embryonic developmental gradient — a non-branching gastrulation→neural-crest axis, no bifurcation. Gast declines while Crest rises to the terminus (smoothed Gast↔Crest r=−0.71). Gate 1 GO, Gate 2 GO (non-branching), Gate 3 NA. The EMT axis is monotone (no root-specific TF; all candidates rise to terminus); ZEB1 is the strongest terminus marker (53% expressing).

Parameter choices

Malignant cells	2,737 raw (18 samples) → threshold cut 125 keeps 6 samples/2,338 → island removal 1 → 2,337	**Expression / features**	log-normalised 10x; 7,498 overdispersed (scFates find_overdispersed); modules vs union control pool
Embedding	n_pcs=20; Harmony on sample (θ=5); kNN k=15; UMAP spread=1.3; one connected component	**Trajectory**	ElPiGraph principal curve on X_umap (Nodes=20, λ=0.3, μ=100); 2 tips, 0 forks; fold-free (self-approach 2.97)
Root / orientation	root = Gast/pSom-high, Crest-low tip; pseudotime left→right (orient_sx=+1, raw coords)	**EMT-TFs**	panel SNAI/ZEB/TWIST + FOXC/PRRX, 3% floor; axis MONOTONE (all rise to terminus); ZEB1 strongest (53%)